

Tandberg TDV-2200 Self-Test Memory Variables

Renamed Memory Variables Used During Diagnostics

Memory Checksum Variables

Address	Old Name	New Name	Purpose	Usage
0x8000	DAT_ram_8000	memory_checksum_byte1	First checksum byte	Sum of these two bytes determines if diagnostics run
0x8001	DAT_ram_8001	memory_checksum_byte2	Second checksum byte	If sum=0 with carry=1, run diagnostics

System Control Variables

Address	Old Name	New Name	Purpose	Usage
0x5E0C	DAT_ram_5e0c	system_state_checksum	System state validation	Determines diagnostic execution, set to 0x8080 after init
0x5E0E	DAT_ram_5e0e	display_control_register	Display control flags	Mirror of port 0x60, controls display modes
0x5E3C	DAT_ram_5e3c	display_mode_register	Display mode settings	Set to 0xBF during initialization

Serial Communication Test Variables

Address	Old Name	New Name	Purpose	Usage
0x5E05	DAT_ram_5e05	serial_test_saved_function_ptr	Backup of serial function pointer	Saves 0x5C2A during serial test
0x5E07	DAT_ram_5e07	serial_test_receive_buffer_flag	Receive buffer status flag	Cleared at test start, checked for errors
0x5E0A	DAT_ram_5e0a	serial_test_mode_flag	Serial test mode selector	Determines test pattern type (0x2D/0x81 vs 0x2E/0x82)
0x5C2A	DAT_ram_5c2a	serial_function_pointer	Active serial function pointer	Modified during serial test to 0x890

Keyboard Controller Test Variables

Address	Old Name	New Name	Purpose	Usage
0x5E01	DAT_ram_5e01	keyboard_test_response_accumulator	Accumulates keyboard responses	Cleared at start, should become 0xAA
0x5E21	DAT_ram_5e21	keyboard_interrupt_flag	Keyboard interrupt occurred flag	Set by keyboard interrupt handler

Display System Variables

Address	Old Name	New Name	Purpose	Usage
0x5E0F	DAT_ram_5e0f	display_sync_control_flag	Display synchronization control	Used in VSYNC processing
0x5E16	DAT_ram_5e16	interrupt_counter	General interrupt counter	Set to 0x101, used for timing
0x5E18	DAT_ram_5e18	display_blink_control_flag	Display element blink control	Controls blinking elements
0x5E19	DAT_ram_5e19	display_cursor_blink_flag	Cursor blink control	Controls cursor blinking
0x5E22	DAT_ram_5e22	display_timing_counter	Display timing counter	Set to 0xA769, used for display timing

Diagnostic Test Flow with Memory Variables

1. System Startup Memory Check

```
assembly
; Check memory integrity
LHLD (memory_checksum_byte1)    ; Load 0x8000-0x8001
ADD H+L                          ; Add bytes together
JNZ/JNC -> run_diagnostics       ; If sum≠0 or no carry, run diagnostics
```

2. System State Validation

```
assembly
; Check system state
LHLD (system_state_checksum)     ; Load 0x5E0C
ADD H+L                          ; Validate system state
CMC                              ; Complement carry
JNZ/JNC -> run_diagnostics       ; Run diagnostics if state invalid
```

3. Serial Communication Test

assembly

```
; Initialize serial test
XRA A
STA (serial_test_receive_buffer_flag) ; Clear 0x5E07

; Save current serial function pointer
LHLD (serial_function_pointer) ; Load 0x5C2A
SHLD (serial_test_saved_function_ptr) ; Save to 0x5E05

; Set test function pointer
LXI HL, 0x890
SHLD (serial_function_pointer) ; Modify 0x5C2A for test

; Test execution with pattern validation
LDA (serial_test_mode_flag) ; Check 0x5E0A for test mode
; Send 0xAA pattern, expect 0xAA back

; Restore original function pointer
LHLD (serial_test_saved_function_ptr) ; Restore from 0x5E05
SHLD (serial_function_pointer) ; Back to 0x5C2A
```

4. Keyboard Controller Test

assembly

```
; Initialize keyboard test
SUB A
STA (keyboard_test_response_accumulator) ; Clear 0x5E01

; Test loop - send 0x80, accumulate responses
; Response handler adds received data to 0x5E01

; Validate final response
LDA (keyboard_test_response_accumulator) ; Load 0x5E01
CPI 0xAA ; Should be 0xAA
JZ success ; Pass if equal
```

Error State Memory Locations

Error Indicators

Condition	Memory State	Description
Serial Buffer Error	serial_test_receive_buffer_flag ≠ 0	Error code 0x15
No Keyboard Response	keyboard_test_response_accumulator ≠ 0xAA	Error code 0x19
Serial Pattern Mismatch	Received data ≠ 0xAA	Error codes 0x17, 0x18
Serial Timeout	No response within timeout	Error code 0x16

Memory Monitoring for Emulator Development

Key Variables to Monitor

```
c
// During serial test
watch 0x5E07 // serial_test_receive_buffer_flag (should stay 0)
watch 0x5E05 // serial_test_saved_function_ptr (backup storage)
watch 0x5C2A // serial_function_pointer (modified during test)

// During keyboard test
watch 0x5E01 // keyboard_test_response_accumulator (should become 0xAA)
watch 0x5E21 // keyboard_interrupt_flag (set by interrupts)

// System state
watch 0x5E0C // system_state_checksum (determines diagnostic execution)
watch 0x8000 // memory_checksum_byte1 + memory_checksum_byte2
```

Emulator Implementation Notes

For Serial Test Success:

- Keep `serial_test_receive_buffer_flag` (0x5E07) = 0
- Echo 0xAA pattern back when written to UART data port
- Ensure `serial_function_pointer` restoration works correctly

For Keyboard Test Success:

- Accumulate 0xAA into `keyboard_test_response_accumulator` (0x5E01)
- Respond to 0x80 command with 0xAA
- Set `keyboard_interrupt_flag` (0x5E21) appropriately

For Memory Checksum Success:

- Set `memory_checksum_byte1` + `memory_checksum_byte2` to non-zero sum with carry
- Initialize `system_state_checksum` (0x5E0C) properly

These renamed variables provide clear insight into the diagnostic test operations and simplify emulator development by showing exactly what memory locations need to be monitored and manipulated.